The Observable Sector from Joint Laws

Blind Selection, Bottom-Up Assembly, and a Certificate-Driven Ladder

Matěj Rada

Mathematical working-paper draft with an executed finite reference battery

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Abstract

We build observable-sector selection from the mature equality-of-law groupoid, not from early placement constructions. The exact observable sector is an algebra of fibre-constant tasks; a tangent quotient and a metric spectral sector are separate later objects. On a finite declared law table, bottom-up assembly closes certified arrows and terminates at the kernel groupoid. A task-aware blind selector returns the complete antichain of inclusion-minimal sufficient protocol panels, retaining ties rather than imposing a hidden preference. Two robustness results distinguish additional observations from additional model uncertainty: protocol aggregation intersects kernels, whereas tasks observable under every model are constant on the equivalence closure of their union. A single model-blind decoder requires the stronger closure of cross-model collisions. We prove these statements and a conditional spectral rank interval, then execute nineteen theorem, counterexample, and control-flow checks. The exhaustive components include 3600 robust-factorization cases, 960 assembly cases, and 5832 common-decoder cases. Reported paired H1–L1 and ECG spectra are reanalysed only as numerical summaries: their 95% variance dimensions are four and three, respectively. The row-level feature tables and original HDF5 records were not recovered in this execution, so no new empirical blind-sector stability or transport test is claimed. The paper closes a finite exact-law selector and a conditional ladder contract; it does not certify a physical source sector from covariance summaries.

1 Three meanings of observable sector

The mature joint-law framework [1, 2] starts with a declared domain X, observation protocols $F_i : X \to Y_i$ whose values are laws, and declared admissible outcome relabelings. Its product-valued law is $F = (F_i)_i$. Equality of these laws defines the kernel groupoid $\mathcal{K}_F$ and the quotient Q_F. Known indistinguishabilities form a bottom-up declared groupoid $A \subseteq \mathcal{K}_F$ when soundness is certified.

There are three different objects that must not be substituted for one another:

1. **Exact observable algebra:** the functions on X constant on $\mathcal{K}_F$-orbits. This is the sector of tasks determined by the exact law.

2. **Regular observable tangent:** $T_xX/\ker DF_x$, under a declared regular differentiable law. It is a quotient, not a preferred horizontal subspace of T_xX.

3. **Metric spectral sector:** an eigenspace selected from a covariance or another operator in a declared feature chart and metric. Its dimension, threshold, and stability depend on that declaration and on estimation error.

The first is exact information. The second needs regularity. The third needs an operator, a metric, a task or threshold, and an error certificate if a statistical lock is requested. A feature covariance is not a source Fisher tensor. A summary spectrum does not determine an eigenspace in a fixed chart.

This work uses the mature groupoid's separation of law-intrinsic vertical directions from protocol-metric observation complements. It does not reuse early placement functionals, a resolved–null coupling as a law invariant, or the identification of local jet isotropy with global law symmetry. Those are explicitly distinguished in the mature source [2].

Execution boundary. The available PDFs include the mature theory, the previous source-automation audit, and the attached ECG report. Attempts to recover the earlier H1/L1 feature CSVs and archive did not supply usable row tables. No ECG feature CSV or waveform was available. Consequently this execution consists of an exact finite-law reference implementation and reported-spectrum arithmetic. It is not a new HDF5 or ECG row-level run. The numerical summaries are inputs, not reconstructed observations.

2 Exact sector and bottom-up assembly

Definition 2.1 (Exact observable algebra)**.** *For a map $F : X \to Y$, let*

$$\mathcal{K}_F = \{(x, x') : F(x) = F(x')\}, \qquad \mathcal{O}(F) = \{T : X \to \mathbb{R} : T(x) = T(x') \text{ on } \mathcal{K}_F\}.$$

The graph of this equivalence relation is the kernel groupoid. Its quotient is canonically $Q_F \simeq \operatorname{im} F$.

The universal factorization theorem says $T \in \mathcal{O}(F)$ if and only if $T = \bar{T} \circ F$ on $\operatorname{im} F$. This is the mature exact layer, not a new claim of physical identifiability [1].

Theorem 2.2 (Bottom-up closure with an exact finite oracle)**.** *Let X be finite, $|X| = n$, and suppose equality of the declared law rows is decidable exactly. Let E be a set of certified arrows with $E \subseteq \mathcal{K}_F$. Its reflexive, symmetric, transitive closure A_E is sound. Repeatedly adding an arrow $(x, x') \in \mathcal{K}_F \setminus A_E$ terminates after at most $n - b$ component-merging additions, where b is the number of $\mathcal{K}_F$-fibres and the initial assembly is the identity. The final orbit relation equals $\mathcal{K}_F$.*

Proof. Equality of laws is an equivalence relation, so closure of arrows inside it remains inside it. Each added arrow joins two different current components and decreases their number by one. Initially there are n components and no sound closure can have fewer than b. If the final relation were a proper subrelation of $\mathcal{K}_F$, a missing within-fibre pair would permit another merge. Thus termination is at b components and equality with $\mathcal{K}_F$. □

With a nontrivial initial sound assembly having c components, the corresponding bound is $c - b$. An unsound initial generator is rejected before closure. The implementation uses union–find and exact rational law rows. Its traversal order is a computational convention; the final equivalence relation is independent of that order. No exact equality oracle is inferred from empirical sample rows.

Proposition 2.3 (Soundness and completeness are distinct)**.** *For a sound assembly $A \subseteq \mathcal{K}_F$, the map $X/A \to Q_F$ is a surjection. It is a bijection exactly when A has the full observation fibres as its orbits. Soundness alone does not certify this.*

Proof. Send the A-orbit of x to its $\mathcal{K}_F$-orbit. Soundness makes this well-defined; surjectivity is immediate. Injectivity is exactly equality of the two orbit partitions. For a constant law on four points, the identity assembly is sound but has four orbits instead of one. □

This is the mature orbit–quotient comparison [2]. It is checked separately rather than inferred from a passing rank calculation.

3 What a blind selector may select

Definition 3.1 (Blindness used here)**.** *The exact selector receives a declared law table, candidate protocols, and, if requested, a declared task T. It does not receive an empirical diagnosis label, a desired rank, or a preferred source representative. It must be equivariant under bijective state and outcome relabeling. If a task or cost is not declared, it may not silently introduce one.*

This is a newly implemented finite reference selector for the mature contract. It is not represented as a recovered historical selector program. A source task in an exact mathematical fixture is a declared function, not an observed clinical label.

Theorem 3.2 (Complete minimal-panel selector)**.** *For a finite protocol index set I, define $F_S = (F_i)_{i\in S}$ and*

$$\mathcal{S}_T = \{S \subseteq I : \mathcal{K}_{F_S} \subseteq \mathcal{K}_T\}.$$

Then $\mathcal{S}_T$ is upward closed. Its inclusion-minimal members form an antichain $\mathcal{M}_T$ and determine it completely:

$$S \in \mathcal{S}_T \quad \Longleftrightarrow \quad \exists M \in \mathcal{M}_T \text{ with } M \subseteq S. \tag{1}$$

Enumerating all panels and retaining all inclusion-minimal sufficient panels is sound and complete. The resulting family is equivariant under bijective relabeling of states, outcomes, and protocol names.

Proof. If $S \subseteq S'$, equality of all laws in S' implies equality of those in S, so $\mathcal{K}_{F_{S'}} \subseteq \mathcal{K}_{F_S}$. Sufficiency is therefore preserved on adding protocols. In a finite upward-closed family every member contains a minimal member, proving (1); minimal members cannot strictly contain one another. Relabeling preserves equality, set inclusion, and task equality, hence carries the whole antichain to the relabeled antichain. □

If $\mathcal{S}_T$ is empty, the candidate protocols do not identify the task. If T is constant, the empty panel is sufficient. If two protocols give the same relevant law, both singleton panels can be minimal. Selecting the first by filename is not a law-forced choice. A declared cost can select cost-minimizers, but ties still remain a correspondence unless another criterion is supplied. Inclusion-minimal is not synonymous with minimum-cardinality or minimum-cost.

Proposition 3.3 (No task-free proper subspace selector)**.** *Let $V = \mathbb{R}^m$ carry its Euclidean metric and the isotropic covariance $C = I$. There is no assignment of a proper nonzero linear subspace $S(C)$ that is equivariant under every orthogonal transformation.*

Proof. Every orthogonal transformation fixes I, so equivariance would make $S(I)$ invariant under all of $O(m)$. If it contains a nonzero vector, rotations carry that vector to every direction and hence $S(I) = V$. Otherwise it is zero. □

A blind spectral selector can retain an entire separated eigenspace. It cannot canonically split a repeated-eigenvalue block. An energy target such as 95% is an operational declaration, not something forced by the bare law or its kernel.

4 Two robustness quotients and a stronger blind-decoder theorem

4.1 More protocols versus more admissible models

Theorem 4.1 (Opposite lattice directions)**.** *On a common latent set X:*

1. *Combining protocols gives*

$$\mathcal{K}_{(F_i)_i} = \bigcap_i \mathcal{K}_{F_i}. \tag{2}$$

2. *For an admissible model family* $\{F_\theta\}$*, let*

$$E_{\mathrm{sep}} = \mathrm{EqCl}\left(\bigcup_\theta \mathcal{K}_{F_\theta}\right).$$

Then

$$\bigcap_\theta \mathcal{O}(F_\theta) = \{T : T \text{ is constant on } E_{\mathrm{sep}}\}. \tag{3}$$

Thus adding observations refines the quotient, whereas requiring observability under more possible models can coarsen the robust task quotient.

Proof. Equation (2) is equality in a product. For (3), constancy on each kernel is equivalent to constancy on their union. Constancy propagates along every finite chain, so it is equivalent to constancy on the equivalence closure of that union. □

Equation (3) says that for every model there is a decoder, possibly a different decoder. It does not yet supply one decoder when the model is unknown. This distinction is essential to the word "blind."

Theorem 4.2 (Common model-blind decoder)**.** *Suppose all* $F_\theta : X \to Y$ *use a common, aligned outcome-law space* Y*. Define the cross-model collision relation*

$$R_\times = \{(x, x') : \exists \theta, \theta' \quad F_\theta(x) = F_{\theta'}(x')\}, \qquad E_\times = \mathrm{EqCl}(R_\times).$$

A single function $g : \bigcup_\theta \operatorname{im} F_\theta \to Z$ *satisfying*

$$T(x) = g(F_\theta(x)) \qquad \text{for every } x, \theta \tag{4}$$

exists if and only if T *is constant on* $E_\times$*. When it exists,* g *is unique on the union of images.*

Proof. If (4) holds, equal cross-model law values have equal target values. Equality of target values extends along chains, proving necessity. Conversely set $g(y) = T(x)$ for any representation $y = F_\theta(x)$. Two such representations give a pair in $R_\times$, so the definition is independent of the representation. Every point of the union has a representation, proving existence and uniqueness there. □

For $X = Y = \{0, 1\}$, take $F_1(x) = x$, $F_2(x) = 1 - x$, and $T(x) = x$. Each kernel is diagonal and the task is observable separately under both models. Yet a common decoder would require $g(0) = 0$ and $g(0) = 1$. Cross-model collisions merge the two states and correctly reject the blind point answer.

These are set-theoretic factorization results. Measurable, continuous, or smooth decoders need the corresponding quotient hypotheses. A common latent domain and aligned outcome meaning are declarations; cross-domain ECG and interferometer spectra do not supply them. No canonical identification between their observable quotients is asserted.

4.2 A coupled law is not just its marginals

For a coupled law P_x^{HL} with marginals P_x^H, P_x^L,

$$\mathcal{K}_{P^{HL}} \subseteq \mathcal{K}_{(P^H, P^L)} = \mathcal{K}_{P^H} \cap \mathcal{K}_{P^L}. \tag{5}$$

Inclusion can be strict because dependence can change while both marginals remain fixed. Equality requires a fixed coupling rule, such as declared conditional independence. This distinction was made explicit in the preceding automation audit [3]. GPS pairing permits contemporaneous record coordinates; it does not specify a full source-indexed coupled law.

5 Finite precision and the metric sector

Theorem 5.1 (Conditional spectral rank interval)**.** *Let $C, \widehat{C}$ be symmetric positive-semidefinite operators in a declared Euclidean chart with $\|C - \widehat{C}\|_{\mathrm{op}} \le \delta$ and $\widehat{\lambda}_1 > \delta$. For a strict relative threshold $\epsilon > 0$, define*

$$r_- = \#\{i : \widehat{\lambda}_i - \delta > \epsilon(\widehat{\lambda}_1 + \delta)\},$$
$$r_+ = \#\{i : \widehat{\lambda}_i + \delta > \epsilon(\widehat{\lambda}_1 - \delta)\}.$$

Then $r_- \le r_\epsilon(C) \le r_+$. If the endpoints agree, the threshold count is certified under the stated bound. Otherwise this certificate is an interval, not a point rank.

Proof. The variational characterization of eigenvalues gives $\widehat{\lambda}_i - \delta \le \lambda_i \le \widehat{\lambda}_i + \delta$. A counted lower-bound direction has $\lambda_i > \epsilon\lambda_1$. Conversely, if $\lambda_i > \epsilon\lambda_1$, then $\widehat{\lambda}_i + \delta > \epsilon(\widehat{\lambda}_1 - \delta)$, so it is included in the upper count. □

The error bound is an input to this theorem. Sweeping a chosen δ is deterministic sensitivity analysis, not an estimated confidence level. A statistical application requires an operator-error certificate under the actual dependence and sampling assumptions. This run supplies no such certificate for either empirical domain.

Proposition 5.2 (Spectra alone do not identify a chart sector)**.** *For $D = \mathrm{diag}(2, 1)$ and a rotation U by $\pi/4$, the matrices D and $UDU^\top$ have identical eigenvalues. Their leading rank-one projectors differ in operator norm by $1/\sqrt{2}$. Therefore an eigenvalue summary cannot identify the leading subspace in a fixed chart.*

Proof. Orthogonal conjugation preserves eigenvalues. The leading vectors have angle $\pi/4$, and direct multiplication of the two rank-one projectors gives norm difference $\sin(\pi/4)$. □

This is non-identification in a fixed chart, not a claim that the two operators are inequivalent under a declared orthogonal gauge. A projector up to an unspecified chart is weaker than a transport map between named channels.

Euclidean covariance energy is also not invariant under arbitrary invertible feature changes. The covariance $\mathrm{diag}(100, 1)$ needs one direction for 95% variance, while the invertible whitening map $A = \mathrm{diag}(0.1, 1)$ sends it to I, which needs two. The observation information has not changed, but the declared Euclidean energy criterion has. A metric must transform with the chart if a geometric invariant is requested.

Finally, $t \mapsto (t, t^2)$ has one-dimensional tangent rank but can have rank-two feature covariance over several t values. The exact fixture $t \in \{-2, -1, 0, 1, 2\}$ exhibits this. A covariance rank cannot be promoted to source tangent dimension or a Fisher-rank certificate.

6 Executed exact-law selector and bottom-up climb

6.1 Declared rational countermodel

The reference model is a mathematical fixture, not synthetic detector data. Let $X = \{(a, b, h) : a, b, h \in \{0, 1\}\}$ and

$$p_a = (1 + 2a)/4, \qquad p_b = (1 + 2b)/4, \qquad c_h = (2h - 1)/32.$$

Protocol H is Bernoulli with success probability p_a, and L is Bernoulli with success probability p_b. Protocol C is the full four-outcome coupled law

$$P_C(a, b, h) = ((1 - p_a)(1 - p_b) + c_h,\ (1 - p_a)p_b - c_h,$$
$$p_a(1 - p_b) - c_h,\ p_a p_b + c_h).$$

All entries are positive rational numbers and sum to one. Its marginals are exactly H, L, while h is encoded only in the dependence. Protocol C identifies all eight declared states.

Protocol panel	Final law fibres	Identity-to-complete orbit counts
H	2	$8, 7, 6, 5, 4, 3, 2$
$H + L$	4	$8, 7, 6, 5, 4$
C	8	8

Within a fixed law, bottom-up closure merges known-indistinguishable states. Across protocol escalation, extra observations can instead split fibres. These are different operations and their opposite count directions are expected.

The task selector returns the following full minimal-panel antichains:

Declared task	Inclusion-minimal sufficient panels
a	$\{H\}$, $\{C\}$
b	$\{L\}$, $\{C\}$
h	$\{C\}$
(a, b)	$\{H, L\}$, $\{C\}$
(a, b, h)	$\{C\}$

No hidden cost says that a coupled protocol is as cheap as a marginal protocol. These are inclusion-minimal panels only. For two identical candidate protocols, the selector retains both singleton alternatives. It does not select one by ordering.

The h-flip assembly is sound and complete for the marginal tuple $H + L$. The same generators change the coupled law C and are rejected as inconsistent there. This demonstrates why a known gauge cannot be transferred to a richer joint law without recertification.

6.2 The nineteen checks

All nineteen modules pass their stated checks and counterexample/sensitivity conditions. The battery includes exact rational/integer fixtures, deterministic matrix examples, seeded relabeling tests, and control-flow checks. It does not estimate a physical false-lock rate.

ID	Checked statement	Attack or scope
T01	Robust task quotient	3600 partition/task cases; intersection mutation
T02	Exact bottom-up assembly	960 partition/edge-subset cases
T03	Soundness is not completeness	Identity assembly, constant law
T04	Coupled law refines marginals	Four marginal versus eight coupled fibres
T05	Relabeling equivariance	100 seeded permutations; garbling is not relabeling
T06	Retain selector ties	Two identical candidate protocols
T07	Selection depends on task	Different targets, different antichains
T08	Gauge depends on protocol	h-flip rejected for coupled law
T09	Garbling enlarges kernel	All 16 binary outcome merges
T10	Spectra do not identify projectors	Isospectral rotation counterexample
T11	Conditional rank interval	101 bounded perturbations; forced point-rank mutation
T12	No proper isotropic selector	Rotation of a chosen coordinate
T13	Tolerance is not a groupoid	Three-point transitivity failure
T14	Ladder dependencies	Kernel pass cannot unlock Fisher
T15	Samples are not exact law oracles	Same finite sample under two positive laws
T16	Covariance rank is not tangent rank	Exact parabola fixture
T17	Common model-blind decoder	5832 map/task cases; model-swap counterexample
T18	Minimal antichain is complete	40 panel/task cases
T19	Metric declaration matters	Whitening changes variance dimension

The sensitivity conditions are not all code-level mutant programs. Some are direct exact counterexamples to a stronger claim; some call an explicitly mutated implementation. Their distinction is visible in the script and JSON. Passing these modules checks the finite implementation and the listed failure modes, not a continuous-law inference engine or a theorem prover for arbitrary experiments.

7 What the available H1/L1 and ECG summaries permit

The paired H1/L1 relative spectrum is taken from the preceding computation in the conversation, consistent with the rounded spectrum in the source-automation PDF [3]. The ECG spectrum is taken from the supplied report [4]. The latter is used only as a numerical input, not as an early placement foundation. No clinical label, old transport conclusion, or placement theorem enters the selector.

The H1/L1 covariance used centered log features without marginal standardization; the ECG report used marginal standardization. Their variance dimensions therefore refer to different declared feature metrics. They cannot be compared as dimensions of one common physical geometry.

Reported summary	$r_{.01}$	$r_{.001}$	$r_{.0001}$	$k_{95\%}$	$k_{99\%}$	$k_{99.9\%}$
Paired H1/L1, 18 coordinates	7	15	18	4	10	15
ECG, 9 coordinates	5	7	8	3	5	7

These are newly executed arithmetic results on supplied eigenvalues. They do not rerun covariance estimation, record deletion, bootstrap stability, held-out reconstruction, or temporal transport. The H1 and L1 individual ranks eight and seven remain prior reported results, not new selector outputs.

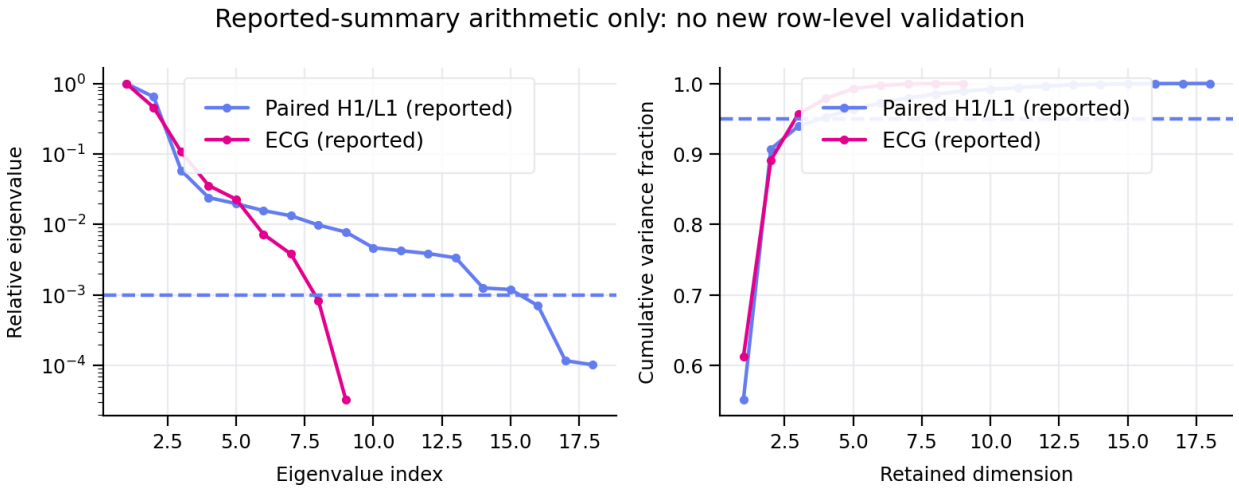


Figure 1: Reported-spectrum arithmetic only. Threshold ranks and cumulative variance dimensions can be computed from eigenvalues. Projector orientation, law sufficiency, confidence, and transport cannot be recovered from this plot.

At $\epsilon = 10^{-3}$, the conditional deterministic rank intervals are:

Assumed $\delta/\widehat{\lambda}_1$	Paired H1/L1	ECG
0	$[15, 15]$	$[7, 7]$
10^{-5}	$[15, 15]$	$[7, 7]$
10^{-4}	$[15, 15]$	$[7, 7]$
10^{-3}	$[13, 18]$	$[7, 9]$
10^{-2}	$[7, 18]$	$[5, 9]$
5×10^{-2}	$[3, 18]$	$[3, 9]$

The displayed spectra are normalized so $\widehat{\lambda}_1 = 1$. No empirical argument establishes any of these operator-error bounds. The table is a sensitivity map, not a statistical confidence statement. In particular, the point intervals at 10^{-4} do not prove a population-rank lock.

The previous HDF5 audit reports injection mask 23 throughout both records, with `NO_CW_HW_INJ` unset. Its explicitly declared all-five-injection-veto policy accepted zero windows [3]. That conservative policy is not asserted to be mandatory for every possible observable task. This run neither relaxes it nor rechecks the raw masks; it retains the earlier failure as provenance. Descriptive spectra are not silently promoted to quality-accepted source samples.

For ECG, the supplied report contains numerical spectra and historical stability summaries but not the row table in this execution. We do not regenerate 200 observations from nine eigenvalues. There is no new ECG label-stratified, patient-level, or clinical inference here.

8 Automated ladder as a dependency graph

The mature ladder's rungs are not a list that must all turn green after one successful computation. The reference climb evaluates named certificate dependencies. The exact finite-law branch can lock its kernel, soundness, completeness, and task descent. It does not have a smooth source family, so Fisher and connection branches remain outside that declared domain.

Requested object	Present execution status	Missing evidence or scope
Finite exact-law kernel	LOCKED	Declared rational fixture only
Finite bottom-up completeness	LOCKED	Exhaustive finite law equality
Finite task-panel antichain	LOCKED	All panels enumerated
Reported-spectrum arithmetic	LOCKED	Supplied numbers only
Empirical blind sector selection	OUTSIDE DOMAIN	Row tables not recovered
Empirical source-law kernel	OUTSIDE DOMAIN	Source law not specified
Chart projector from spectrum	CORRESPONDENCE	Eigenvectors not supplied
Statistical rank certificate	Not certified	No operator-error bound
Regular quotient/Fisher	OUTSIDE DOMAIN	No regular source law derivative
Observable connection	OUTSIDE DOMAIN	No regular metric field and path data
Latent lift	OUTSIDE DOMAIN	No declared latent covariance
Physical scale	OUTSIDE DOMAIN	No relevant anchor

The projector correspondence is an isospectral family in a fixed chart, not a claim that a full source-model correspondence has been computed. The source-model request is outside the supplied domain. Distinguishing these two abstentions prevents a summary-level result from masquerading as a physical inverse solution.

Theorem 8.1 (Adaptive selection requires simultaneous certificates)**.** *Let a declared candidate collection have gate-failure events E_j with $\Pr(E_j) \leq \alpha_j$. If every reported lock, including a*

data-adaptively selected one, is valid whenever none of the E_j occurs, then

$$\Pr\{\textit{any reported false lock}\} \le \sum_j \alpha_j.$$

For a countable adaptive ladder, the same statement holds when the error budget is summable and each event has the stated bound.

Proof. A false lock implies at least one gate failure, so its event is contained in $\bigcup_j E_j$. Apply the union bound, using countable subadditivity in the countable case. □

This does not supply the gate probabilities. It explains why selecting a favorable panel and then attaching an unadjusted per-panel confidence claim is not automatically valid. An empirical blind selector would need a frozen candidate/task contract, dependence-aware certificates or a justified holdout design, and an error budget for its escalation. None is inferred from the summary spectra.

9 Final theorem package and limits

The supported theorem package is:

1. Exact observable tasks are functions on the joint-law quotient; sound bottom-up assembly reaches it under a complete exact finite oracle.

2. A finite task-aware blind selector can return the complete minimal sufficient-panel antichain without a hidden tie-breaking preference.

3. Adding protocols intersects kernels. Requiring separate observability under all admissible models takes the equivalence closure of their union. A common model-blind decoder requires the stronger cross-model collision closure.

4. A declared operator-error bound yields a rigorous relative-rank interval. Spectral arithmetic alone does not identify a chart projector or certify a source tangent.

5. Certificate dependencies and simultaneous error accounting prevent a lower-rung pass or an adaptively chosen candidate from silently unlocking a higher structure.

These are conditional mathematical statements with proofs. The finite execution checks them and their named counterexamples; it does not establish universality from two measurement domains.

The empirical part is intentionally narrower: two reported spectra are reanalysed, not two raw datasets rerun. The requested row-level blind selector and automated empirical stability climb are not completed because the needed tables were not recovered. No new empirical observable-sector theorem, source reconstruction, clinical conclusion, law-natural latent connection, or cross-domain geometry identity is claimed.

The resulting paper is therefore a finalized mathematical and executable reference contract, with the empirical validation boundary left explicit. It is not a finalized empirical lock on the physical observable sector of the H1/L1 or ECG source systems.

A Reproducibility

The script `rg_observable_climb.py` runs with Python and NumPy:

```
python rg_observable_climb.py -out-dir .
```

It emits `rg_observable_climb_results.json`, including the exact rational fixture, minimal-panel antichains, bottom-up orbit-count traces, all nineteen check records, reported-spectrum arithmetic, package versions, input-PDF hashes, script hash, and an explicit data-availability ledger. It does not download data or fabricate missing rows. A separate plot script reproduces the summary figure.

The four-point exhaustive assembly check enumerates all 15 set partitions and all 64 subsets of the six possible undirected edges: 960 cases. The robust-algebra check enumerates all 225 pairs of partitions and all 16 binary tasks: 3600 cases. The common-decoder check enumerates all 27 maps from a three-point set to a three-point set, all 729 ordered map pairs, and all eight binary tasks: 5832 cases. These counts are generated by the execution, not sample-size claims about detectors or patients.

No external web search was used. Web search was unavailable; the background knowledge cutoff is December 2025. The theoretical grounding is the supplied mature working papers, and all new proofs are stated here. Current external scientific or bibliographic verification is not claimed.

References

[1] M. Rada. *Resolution Geometry: The Theorem Ladder*. Supplied September 2026 mature working paper, 24 pages. Exact layer, regularity, natural descent, rank certificates, and closure.

[2] M. Rada. *The Groupoid of Resolution: From the Kernel Pair to Global Transport, Bottom-Up and Top-Down*. Supplied October 2026 working paper, 8 pages.

[3] M. Rada. *Resolution Geometry: Source-Side Automation. Joint-Law Descent, Certified Correspondences, and a Fail-Closed H1–L1 HDF5 Audit*. Previous October 16, 2026 computation and supplied 9-page PDF. Used for provenance and reported spectral inputs.

[4] M. Rada. *Cross-Domain Observable-Sector Admissibility in Real 12-Lead ACS-ECG Data*. Supplied July 31, 2026 report, 5 pages. Used only for its printed numerical spectrum and data-availability context, not as a placement foundation.